



Department of Physics

National Institute of Technology Kurukshetra

Course: B.Tech (EE, SET, CE, ECE, ME, PIE, R&A)

Exam: Sessional-II

Date: 23-10-2024

Subject: Engineering Physics

Course Code: PHIC 101

Duration: 50 min

Note: All questions are compulsory.		Max. Marks: 20	Roll No:
1.	Which of the following is a most appropriate wavefunction and why? (i) $\psi = A \sin x$, (ii) $\psi = A \exp(x^2)$, and (iii) $\psi = A \exp(-x^2)$.	5	
2.	(a) What is Ampere's law? Why Maxwell amended the Ampere's law? Discuss the significance of Maxwell-Ampere's law. (b) Write down the Four Maxwell's equations in a conducting medium.	3+2	
3.	Draw the following Miller planes; (1 1 0), (6 3 4), (3 2 $\bar{1}$), ($\bar{3}$ 0 $\bar{4}$) and (1 $\bar{2}$ 1).	5	
4.	(a) In the Drude model, electrons are treated as classical particles. Explain why this assumption works well for some properties of metals but fails for others. (b) The conductivity of aluminium at room temperature is $3.5 \times 10^7 \text{ S/m}$, and the density of free electrons is $6 \times 10^{28} \text{ m}^{-3}$. Calculate the electron mobility (μ) in aluminium.	3+2	

$$\frac{-qE}{\sqrt{3k_B T_m}} \times \frac{\lambda_m}{v_d} = \frac{\lambda}{\tau}$$

$$J = ne v_d$$

$$J = \frac{q^2 E \lambda}{\sqrt{3k_B T_m}}$$

$$J = \frac{1}{\mu_0} \frac{d\vec{B}}{dt} = \frac{1}{\mu_0} \frac{d}{dt} \int \vec{B} \cdot d\vec{l}$$

$$\vec{E} = -\frac{d\phi}{dt} \quad \phi = \oint \vec{E} \cdot d\vec{s}$$

$$\oint \vec{E} \cdot d\vec{l} = \frac{q}{\epsilon_0} \quad \vec{E} = \frac{q}{\epsilon_0} \frac{\vec{r}}{r^2}$$